Review of OLG Chapter Revision

Errors and Suggested Improvements in `black8_ver4_Lars.tex`

Review Notes

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1 English Grammar and Typographical Errors

G1. [p. 404, §10.1, TeX line ∼144] **Typo: "inital" → "initial."**
The text reads: "the **inital** old consumer $i = -1\ldots$"
Should be: "the **initial** old consumer $i = -1\ldots$"

G2. [p. 406, §10.3, TeX line ∼213] **Missing word: "lets analyze" → "lets us analyze."**
The text reads: "this equilibrium concept **lets** analyze a richer set…"
Should be: "this equilibrium concept **lets us** analyze a richer set…"

G3. [p. 409, §10.3.3, TeX line ∼494] **Typo: "equilibrum" → "equilibrium."**
The text reads: "equilibrium L is also a Walrasian **equilibrum**."
Should be: "…a Walrasian **equilibrium**."

G4. [p. 413, §10.3.5, TeX line ∼710] **Imprecise term: "coordinate axis" should be "abscissa" or "horizontal axis."**
The text reads: "y^t_{t+1} and both c^t_{t+1} and c^{t-1}_t are on the **coordinate** axis."
The vertical axis is the *ordinate* and the horizontal axis is the *abscissa*. Suggest: "…are on the **horizontal** axis."

G5. [p. 421, §10.6, TeX line ∼1195] **Mismatched parenthesis: "real ('indexed) bonds."**
The text reads: "…by issuing real **('indexed)** bonds."
Should be: "…by issuing real **(indexed)** bonds." (or "real, i.e., indexed, bonds").

G6. [p. 429, §10.7.2, TeX line ∼1632] **Sentence fragment and spelling.**
The text reads: "…will converge to zero**, Though, as we have learnt,**…"

- "Though" begins a sentence fragment; should be restructured.
- "learnt" is British; the rest of the chapter uses American spelling.

Suggested fix: "...will converge to zero**. However, as we have learned,**..."

G7. [p. 430, §10.7.2, TeX line ∼1645] **Ungrammatical infinitive.**
The text reads: "So what makes the present value budget constraint **to hold**?"
Should be: "So what makes the present value budget constraint **hold**?"

G8. [p. 432, §10.8.1, TeX line ∼1775] **Typo: "governement" → "government."**
The text reads: "the **governement** finances deficits..."
Should be: "the **government** finances deficits..."

G9. [p. 434, §10.8.2, TeX line ∼1860] **Ungrammatical sentence.**
The text reads: "**Disliking the stability of the bad Laffer curve equilibrium arises under perfect foresight dynamics.**"
The gerund "Disliking" as the subject makes the sentence incoherent.
Suggested fix: "**The undesirable stability of the bad Laffer-curve equilibrium is a feature of perfect foresight dynamics.**"

G10. [p. 434, §10.8.2, TeX lines ∼1830, 1855] **Editorial placeholders remain in published text.**
The text contains bracketed notes:

- "**[Elaborate]**"
- "**[Elaborate on this. Then connect back to graphs...]**"

These should be removed or expanded upon.

G11. [§10.9.2, TeX line ∼1880] **Missing space: "someissues" → "some issues."**
The text reads: "to think about **someissues** to be discussed..."
Should be: "to think about **some issues** to be discussed..."

G12. [p. 406, §10.3, TeX lines ∼200–210] **Inconsistent possessive: "Walras Law" vs. "Walras' Law."**
Both forms appear within a few lines of each other. Pick one consistently; the standard in economics is "Walras' law" or "Walras's law."

2 Mathematical and Logical Errors

M1. [p. 415, §10.3.5, TeX line ∼850] **Wrong equation reference for nonstationary dynamics.**
The text reads: "To initiate the computation of a nonstationary equilibrium using difference equation **(10.3.13)**..."
But equation (10.3.13) is the *stationary* quadratic $\alpha^2 - \epsilon^{-1}\alpha + (\epsilon^{-1} - 1) = 0$. The nonstationary dynamics are governed by the *difference equation* (10.3.12):

$$\alpha_i = \epsilon^{-1} - \frac{\epsilon^{-1} - 1}{\alpha_{i-1}}.$$

The reference should be to **(10.3.12)**, not (10.3.13).

M2. [p. 419, §10.5.1, TeX line ∼1125] **Ambiguous parsing of inline division.**
Equation (10.5.9) is typeset as:

$$p_t = 2M/w_1 + \left(\frac{w_2}{w_1}\right) p_{t+1}.$$

The term $2M/w_1$ could be misread. It should be parsed as $(2M/w_1) + (w_2/w_1)\, p_{t+1}$. Adding parentheses or using a display fraction would improve clarity.

M3. [§10.10, TeX line ∼2136] **Wrong variable name: "c" should be "κ."**
The text reads: "recall the continuum of equilibria indexed by the scalar $c > 0$ in expression (10.5.10)."
But in equation (10.5.10) the scalar is called κ, not c.
Should read: "...indexed by the scalar $\kappa > 0$..."

M4. [TeX lines ∼668 and ∼1744] **Duplicate equation label `equilib1;a`.**
The TEX label `equilib1;a` is used twice:

- First, for the offer-curve equation (10.3.9a) at line ∼668.
- Second, for the money-financed equilibrium definition (10.8.2a) at line ∼1744.

This will cause a cross-referencing error. The second usage should be given a distinct label (e.g., `Meq_def;a`).

3 Inconsistencies Across Sections

I1. [§10.9 heading, TeX line ∼1899] **Placeholder section title.**
Section 10.9 is titled "**Remain to be revised**." This should be removed or replaced with a proper section heading.

I2. [§§10.9.1–10.9.2, TeX lines ∼1905–1960] **Old time-indexing convention.**
Sections 10.1–10.8 (the revised portion) consistently start time at $t = 0$, with the initial old labeled $i = -1$. Sections 10.9.1 ("Another Gale diagram") and 10.9.2 ("Endowment at $+\infty$") still use the *old* convention where time starts at $t = 1$ and the initial old are $i = 0$. For example:

- "the budget constraint of an initial old person... $q_1^0 c_1^0 = \ldots$"
- "$y_1^0 = \epsilon > 0$"

These sections need their indexing updated to match the revised conventions ($t \geq 0$, initial old $i = -1$).

I3. [§10.10, TeX line ∼2028] **Switch from one-argument to two-argument utility without notice.**
Sections 10.1–10.8 define $u(c)$ as a one-argument per-period utility function and write lifetime utility as $u(c_i^i) + u(c_{i+1}^i)$. Section 10.10 switches without comment to a two-argument utility $u(c_t^t, c_{t+1}^t)$.
Suggested fix: add a transitional sentence, e.g., "In this section, we generalize by writing the agent's utility function as $u(c_1, c_2)$, a function of both young and old consumption."

I4. [§10.10, TeX line ∼2050] **Time starts at $t = 1$ in §10.10, not $t = 0$.**
This section says "The welfare of the initial old agents at time 1" and the economy starts at period 1. This is inconsistent with sections 10.1–10.8, where time starts at $t = 0$. If this section is retained, the time convention should be harmonized.

I5. [p. 430, §10.7.2, TeX line ∼1596] **Potentially confusing sign convention.**
The text writes: "the consumption of the initial old is $c_0^{-1} = w_2 - \tau_0^{-1}$." Since $\tau_0^{-1} \leq 0$ (it is a tax/transfer with $-\tau_0^{-1} \geq 0$ being the transfer), a reader might misread $w_2 - \tau_0^{-1}$ as making consumption *smaller*. A brief clarifying note such as "(recall $\tau_0^{-1} \leq 0$, so $-\tau_0^{-1} \geq 0$)" would help.

4 Summary of Most Critical Items

Table 1 collects the highest-priority issues.

#	Type	Location	Issue
G2	Grammar	p. 406, §10.3	"lets analyze" → "lets us analyze"
M1	Math ref	p. 415, §10.3.5	Wrong eqn ref: should cite (10.3.12) not (10.3.13)
M3	Math	§10.10	Scalar called "c" but should be "κ"
M4	TEX	Two locations	Duplicate equation label `equilib1;a`
G1	Typo	p. 404, §10.1	"inital" → "initial"
G8	Typo	§10.8.1	"governement" → "government"
G3	Typo	§10.3.3	"equilibrum" → "equilibrium"
G9	Grammar	§10.8.2	"Disliking the stability... arises" — ungrammatical
G10	Draft	§10.8.2	[Elaborate] placeholder notes remain
I1	Draft	§10.9	Section title "Remain to be revised"
I2–I4	Consistency	§§10.9–10.10	Old time-indexing and notation

Table 1: Summary of highest-priority issues.